

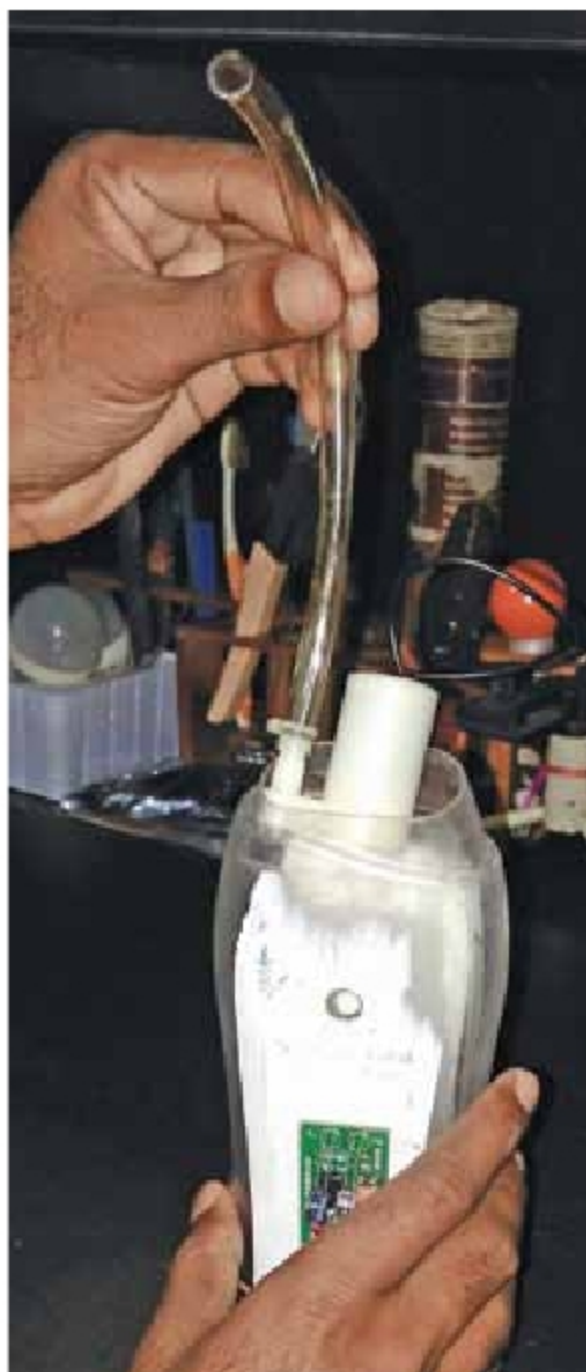


Fig. 4: Insert DC pump and pipe inside the bottle

de-energises relay. When the submersible pump disconnects from the power supply, the DC pump goes off, which stops the flow of hand sanitiser.

Construction and testing

You can use a 5V DC adaptor to operate the circuit. Assemble the circuit on a general-purpose PCB and enclose it in a suitable cabinet. Attach the IR sensor module on the front of the dispenser bottle. The proposed construction details are explained here:

1. The author has used a shampoo bottle for this project. First, connect various components used in the project, as shown in Fig. 3.

2. Trim shampoo bottle at a suitable length. Insert DC pump and small plastic pipe (20-30cm) inside the bottle, as shown in Fig. 4.

3. Make a small hole on the bottle so that one end of the plastic pipe



Fig. 5: Fitting the plastic pipe on the bottle

can be fitted on the bottle, as shown in Fig. 5.

4. Attach the remaining components, including the main circuit and IR module, on the bottle using glue. Pour down sanitiser into the bottle and cover it with an airtight lid. The final proposed assembly is shown in Fig. 6.

Switch on the circuit and keep your hands in front of the IR sensor module. Sanitiser fluid will flow out and continue to flow out until your hands are withdrawn from the sensor.

You need to cover the IR module with suitable insulation and waterproof material so that it would not get damaged in case liquid falls on the module. The IR sensor detects hand at a distance in the range between 2 ~ 30cm. You can manually calibrate the sensitivity using an inbuilt potentiometer on the IR module.

Note. The IR module tends to get activated by itself when kept in open



Fig. 6: Final proposed assembly of hand sanitiser dispenser

PARTS LIST

Semiconductors:

T1	- BC547 npn transistor
D1	- 1N4007 rectifier diode
LED1	- 5mm LED
IR1	- 3-pin IR sensor module (LM358 based)

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R2	- 1-kilo-ohm
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Miscellaneous:

RL1	- 5V single changeover relay
DC pump	- Mini DC (3V-5V) submersible pump
CON1, CON3,	
CON4	- 2-pin male Berg connector
CON2	- 3-pin male Berg connector
	- 5V DC adaptor

sunlight and sometimes even under tubelight inside a room. It is recommended to use a suitable contraption or enclosure for the IR module to prevent false triggering by ambient light. **EFY**



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